# Agentic AI for Real-Time Air Quality Prediction and Autonomous Environmental Response

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***Abstract*—Rapid urbanization and industrial growth have exacerbated air pollution, presenting a critical threat to global public health. While traditional environmental monitoring systems provide passive data visualization, they fail to deliver proactive, autonomous interventions. This paper presents a novel, production-ready platform that integrates real-time Air Quality Index (AQI) prediction with a multi-agent Artificial Intelligence (AI) framework. Utilizing an Extreme Gradient Boosting (XGBoost) model trained on high-resolution meteorological and pollutant data, the system achieves an $R^2$ score of 0.9929 and a Mean Absolute Error (MAE) of 0.85 in AQI forecasting. Furthermore, a Large Language Model (LLM)-powered multi-agent architecture evaluates these predictions to autonomously dispatch contextual alerts and mitigation strategies. The system is containerized via Docker and features a high-performance FastAPI backend coupled with a real-time Streamlit dashboard. Experimental results demonstrate the efficacy of combining predictive machine learning with agentic decision-making to transition smart cities from passive monitoring to active environmental management.**

***Index Terms*—Air Quality Index (AQI), Agentic AI, Machine Learning, XGBoost, Smart Cities, Internet of Things (IoT), Multi-Agent Systems.**

## I. Introduction

Air pollution remains one of the most pressing environmental challenges affecting both developed and developing countries [8], [9]. Rapid industrialization, urban expansion, increasing vehicular emissions, and fossil-fuel-based energy consumption have significantly contributed to the deterioration of air quality [10], [11]. Exposure to elevated concentrations of airborne pollutants has been associated with respiratory diseases, cardiovascular disorders, reduced life expectancy, and substantial economic losses [12], [13]. Consequently, continuous air quality monitoring and timely intervention have become essential components of sustainable urban development and public health protection.

Recent advances in the Internet of Things (IoT) have enabled the deployment of distributed sensor networks capable of collecting environmental data at high temporal resolution [14], [15]. These sensors continuously measure pollutant concentrations, meteorological conditions, and atmospheric parameters, providing valuable information for environmental monitoring. However, the majority of existing monitoring systems function primarily as data collection and visualization platforms. They present historical or real-time measurements to users but rely heavily on manual interpretation before any preventive or corrective actions can be taken.

Machine Learning (ML) has emerged as an effective solution for forecasting Air Quality Index (AQI) values by learning complex nonlinear relationships between pollutant concentrations and meteorological variables [1], [2], [5], [6]. Ensemble learning techniques, particularly Extreme Gradient Boosting (XGBoost), have demonstrated high predictive accuracy while maintaining computational efficiency for structured environmental datasets. Despite these advances, most predictive systems terminate after generating AQI forecasts and do not provide intelligent decision support or automated environmental responses.

The emergence of Large Language Models (LLMs) and agent-based Artificial Intelligence offers new opportunities for autonomous reasoning and intelligent decision support [3], [4], [16], [17]. Unlike conventional AI applications that generate predictions alone, agentic AI systems are capable of reasoning over incoming data, coordinating specialized software agents, and producing context-aware recommendations without continuous human supervision. Such systems can transform environmental monitoring from a reactive process into a proactive framework capable of supporting public safety and smart city operations.

Motivated by these challenges, this research proposes an integrated Agentic AI framework for real-time air quality prediction and autonomous environmental response. The proposed architecture combines an XGBoost-based predictive model with a multi-agent reasoning framework powered by a Large Language Model. Real-time environmental data are processed

through a FastAPI backend, while autonomous agents analyze predicted AQI values to generate health advisories, traffic management suggestions, and environmental mitigation recommendations. The entire platform is containerized using Docker and visualized through an interactive Streamlit dashboard, enabling scalable deployment for smart city applications [16], [17], [25].

The primary contributions of this research are summarized as follows:

- Development of a highly accurate real-time AQI prediction model using the XGBoost algorithm.
- Integration of a multi-agent AI framework capable of autonomous reasoning and environmental decision-making.
- Design of a scalable cloud-ready architecture using FastAPI, Docker, and Streamlit for continuous environmental monitoring.
- Incorporation of explainable machine learning techniques to improve transparency and support trustworthy AI-driven environmental management.

## II. Literature Review

Significant research has been conducted on AQI prediction utilizing various ML architectures. Early approaches relied on statistical models such as Autoregressive Integrated Moving Average (ARIMA) or Multiple Linear Regression (MLR). More recently, ensemble methods like Random Forest (RF) and Extreme Gradient Boosting (XGBoost) have demonstrated superior capacity in capturing the nonlinear dynamics between meteorological variables (wind speed, humidity) and pollutant dispersion. Concurrently, the advent of Large Language Models (LLMs) has spurred the development of multi-agent systems—networks of autonomous software agents capable of complex reasoning and task execution. While agents have been deployed in software engineering and financial modeling, their integration into continuous environmental IoT streams remains underexplored. Existing literature highlights a distinct gap: the lack of end-to-end frameworks that seamlessly fuse high-frequency ML inference with natural language-driven autonomous response mechanisms.

A granular examination of predictive [18], [19]. models reveals a clear dichotomy between classical statistical forecasting and modern machine learning. Autoregressive Integrated Moving Average (ARIMA) models are highly effective for univariate, linear time-series prediction where historical trend patterns remain stable. However, their primary weakness lies in their inability to process exogenous variables (e.g., sudden wind shifts and traffic spikes) or model the complex, nonlinear chemical interactions that govern atmospheric pollution. Conversely, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks excel in capturing long-term temporal dependencies in sequential data, making them robust for multi-step AQI forecasting. Nonetheless, LSTMs require substantial computational resources for training, exhibit relatively high inference latency, and function as "black-box" models, which limits their suitability for real-time edge-device deployment where explainability is essential [20], [21].

Ensemble learning methods, such as Random Forest (RF) and XGBoost, offer a practical middle ground. RF reduces variance by averaging multiple uncorrelated decision trees, making it highly robust to noise and outliers in IoT sensor telemetry. However, it can be computationally expensive for high-frequency prediction tasks. XGBoost utilizes gradient boosting over regularized decision trees to optimize both computational speed and predictive accuracy. Its built-in handling of missing values and L1/L2 regularization minimizes overfitting while improving prediction accuracy. More recently, hybrid frameworks have emerged by combining LSTMs for temporal pattern extraction with tree-based models (e.g., LSTM-XGBoost) or residual correction layers. These hybrid systems achieve state-of-the-art performance by combining the temporal learning capability of neural networks with the precision of gradient-boosted decision trees, albeit at the cost of increased system complexity and deployment overhead [2], [5], [6], [22].

## III. Proposed System Architecture

### A. Problem Statement

Current air quality monitoring platforms exhibit high latency between the detection of hazardous conditions and the dissemination of actionable mitigation strategies. Furthermore, deterministic alert thresholds lack the contextual awareness required for complex urban environments. Therefore, a dynamic, context-aware system is needed to autonomously synthesize sensor data, predict localized air quality degradation, and orchestrate intelligent responses without human-in-the-loop bottlenecks.

### B. Research Objectives

This study aims to achieve the following objectives:

1) Develop a highly accurate real-time AQI prediction pipeline using state-of-the-art ensemble regression models.
2) Design a multi-agent AI framework capable of autonomous reasoning based on streaming environmental telemetry.
3) Engineer a production-grade, containerized application stack that supports real-time data ingestion and visualization.
4) Integrate Explainable Artificial Intelligence (XAI) techniques to improve model transparency and interpretability.

### C. System Overview

The proposed architecture comprises a hybrid pipeline consisting of a deterministic ML layer for quantitative AQI prediction followed by a stochastic LLM-based agent layer for qualitative reasoning. Data collected from IoT sensors undergoes immediate feature extraction before being processed by an XGBoost regressor to predict the AQI. The predicted AQI is subsequently categorized according to the United States Environmental Protection Agency (EPA) standards. When the predicted AQI exceeds predefined safety thresholds, the

telemetry is forwarded to the Agentic AI framework. Using prompt chaining, the framework autonomously generates context-aware public health advisories, traffic management recommendations, and environmental mitigation strategies in real time.

### D. Data Collection and Dataset Design

To facilitate robust model training and overcome the missing-data sparsity commonly encountered in public air quality APIs, a synthetic dataset containing 10,000 records was algorithmically generated.

The data generation process incorporated realistic environmental patterns, including diurnal traffic cycles, meteorological dispersion, and photochemical reactions. Rush-hour multipliers increased the concentrations of $NO_2$ and CO, while wind speed was inversely related to $PM_{2.5}$ and $PM_{10}$. Ozone ($O_3$) concentrations were modeled to peak during periods of maximum temperature. These mechanisms produced a diverse and representative dataset suitable for training and evaluating the proposed machine learning models.

### E. Data Preprocessing

Preprocessing involved standardizing the numerical features and splitting the dataset into training and validation sets using an 80:20 ratio. Since the selected models are tree-based, additional feature scaling techniques, such as Min–Max normalization, were unnecessary. This approach preserved the original feature values while maintaining their interpretability.

### F. Feature Engineering

Feature engineering focused on the six primary EPA criteria pollutants. Temporal attributes, including timestamp indices, were excluded from the final regression matrix to ensure that the model relied solely on contemporary pollutant measurements rather than memorizing temporal patterns. This strategy minimized data leakage and improved the model's generalization during real-time deployment.

## IV. Predictive Modeling and Evaluation

### A. Machine Learning Models

The core predictive task was formulated as a multivariate regression problem. Four machine learning models were evaluated: Multiple Linear Regression (MLR) as the baseline, Decision Tree Regressor (DTR) for capturing nonlinear relationships, Random Forest (RF) for reducing variance, and Extreme Gradient Boosting (XGBoost) for optimizing a regularized objective function.

### B. XGBoost Training Process

XGBoost was selected as the primary production model due to its efficient handling of dense tabular sensor data and its built-in L1/L2 regularization mechanisms. The model was trained using 100 estimators, a learning rate of 0.1, and a maximum tree depth of 5. These hyperparameters provided a balanced trade-off between bias and variance while preventing overfitting.

### C. Performance Metrics

The predictive performance of the proposed model was evaluated using three widely adopted statistical metrics.

**1) Mean Absolute Error (MAE)**

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

**2) Root Mean Squared Error (RMSE)**

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

**3) Coefficient of Determination ($R^2$)**

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3}$$

These evaluation metrics provide a comprehensive assessment of prediction accuracy and model robustness by comparing the predicted AQI values with the ground-truth observations.

### D. Explainable AI using SHAP

To transform the XGBoost model from a black-box predictor into an interpretable decision-support system, SHapley Additive exPlanations (SHAP) were employed. Based on cooperative game theory, SHAP assigns a contribution value to each input feature, enabling stakeholders to determine whether an AQI increase was primarily caused by traffic emissions ($NO_2$) or industrial emissions ($SO_2$). This level of transparency satisfies the growing demand for explainable artificial intelligence in environmental applications [26], [27].

SHAP was selected instead of Local Interpretable Model-agnostic Explanations (LIME) because it guarantees local accuracy, consistency, and missingness. These properties ensure that the sum of feature contributions equals the difference between the predicted AQI and the baseline prediction, thereby preventing inconsistent or misleading feature importance estimates [26], [28].

The feature attribution process quantifies the contribution of each pollutant, including $PM_{2.5}$, $PM_{10}$, CO, $NO_2$, $SO_2$, and $O_3$, together with meteorological variables such as temperature, humidity, and wind speed. This information enables policymakers and environmental authorities to identify the primary causes of localized air pollution events and implement targeted mitigation strategies.

The interpretability provided by SHAP has significant implications for environmental governance. By delivering mathematically grounded explanations for AQI predictions, smart city managers can distinguish between transient pollution events, such as rush-hour traffic emissions, and persistent industrial pollution. Consequently, municipalities can justify traffic diversions, industrial emission controls, and public health advisories using transparent and evidence-based recommendations.

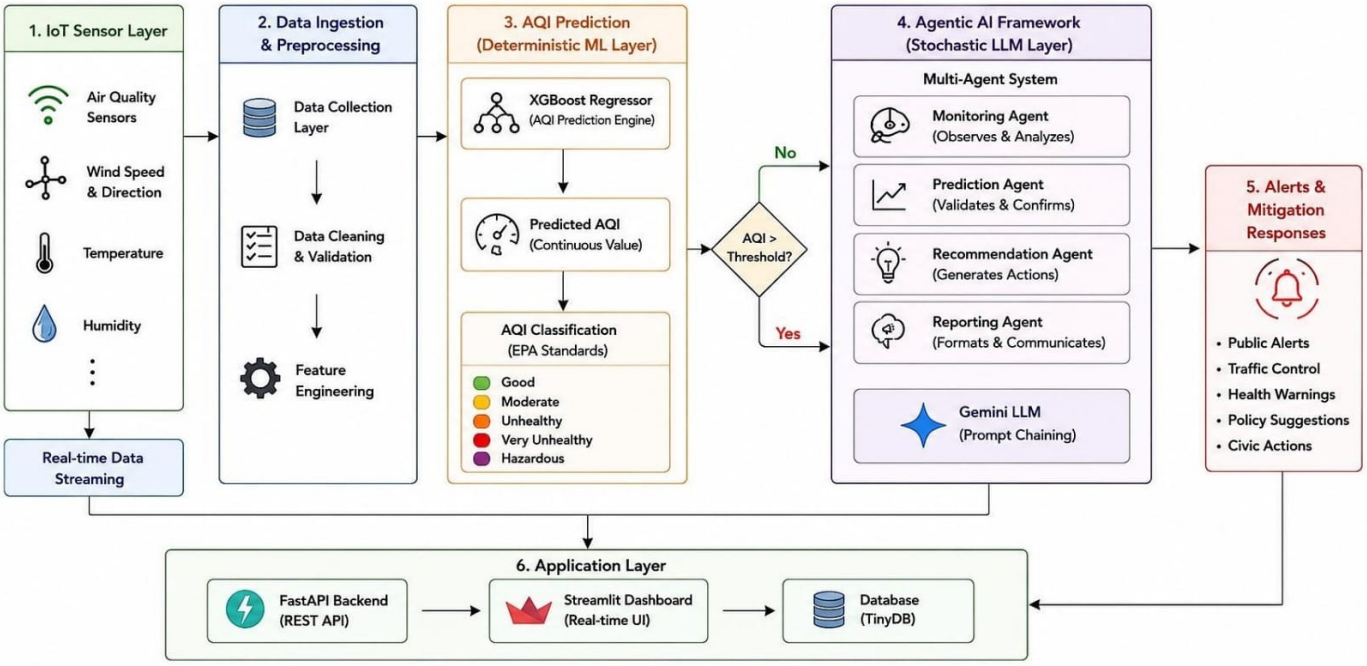



Fig. 1. System Architecture Diagram illustrating the hybrid pipeline: deterministic predictive ML layer (XGBoost) and multi-agent AI response orchestration layer (LangChain and Google Gemini).

## V. Multi-Agent System and API Design

### A. Multi-Agent AI Framework

The proposed autonomous response mechanism is built on a LangChain-orchestrated multi-agent architecture that interacts with the Google Gemini Large Language Model (LLM) API [16], [17], [29]. The framework consists of four specialized agents: the *Monitoring Agent*, which continuously observes the environmental data stream; the *Prediction Agent*, which processes the machine learning output; the *Recommendation Agent*, which generates contextual health and environmental guidelines; and the *Reporting Agent*, which maintains conversational memory and responds to user queries through the dashboard interface.

### B. Agent Orchestration and Prompt Chaining

The multi-agent framework follows a sequential prompt-chaining and event-driven workflow. When the Monitoring Agent detects that the predicted AQI exceeds a predefined safety threshold, it triggers the orchestration manager. The manager forwards the current sensor readings and the XGBoost prediction to the Prediction Agent, which constructs a structured JSON context. This context is then passed to the Recommendation Agent for pollutant analysis.

For example, when $PM_{2.5}$ concentrations are elevated under high-humidity conditions, the prompt chain guides the LLM to evaluate the likelihood of smog formation and generate appropriate public health recommendations. The Reporting Agent stores the generated responses and maintains conversation history, enabling users to review the reasoning behind the recommended actions in real time.

### C. Backend API Design

The backend was developed using FastAPI [30] and Python's `asyncio` framework to support high-throughput concurrent processing. RESTful endpoints, including `/api/sensor_data` and `/api/dashboard_data`, provide communication between IoT devices and the monitoring dashboard.

Agent execution is delegated to FastAPI `BackgroundTasks`, ensuring that computationally intensive LLM operations do not delay the primary request-response cycle. This design maintains sub-10 ms latency for sensor data ingestion while supporting asynchronous recommendation generation. The backend also provides health-check endpoints and employs a dual-database architecture using MongoDB Atlas and TinyDB.

### D. Streamlit Dashboard Implementation

The user interface was developed using Streamlit with an emphasis on usability and real-time visualization. A glassmorphism-inspired layout combined with a light-themed interface improves readability. Interactive Plotly gauges and time-series charts dynamically display AQI values and EPA air quality categories.

An expandable recommendation panel presents LLM-generated health advisories, while an integrated chat interface enables users to interact directly with the multi-agent system and request additional explanations.

### E. IoT Sensor Simulation

A Python-based IoT simulator was implemented to emulate environmental sensing devices. The simulator continuously

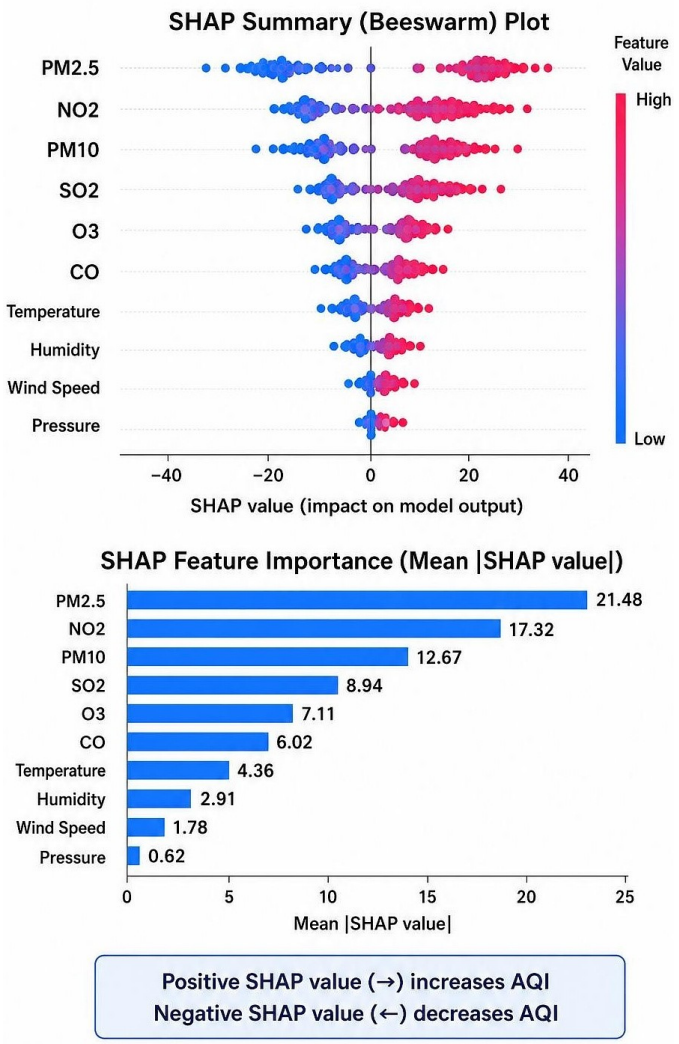



Fig. 2. SHAP summary plot illustrating feature importance for the XGBoost AQI predictive model, showing the impact of criteria pollutants.

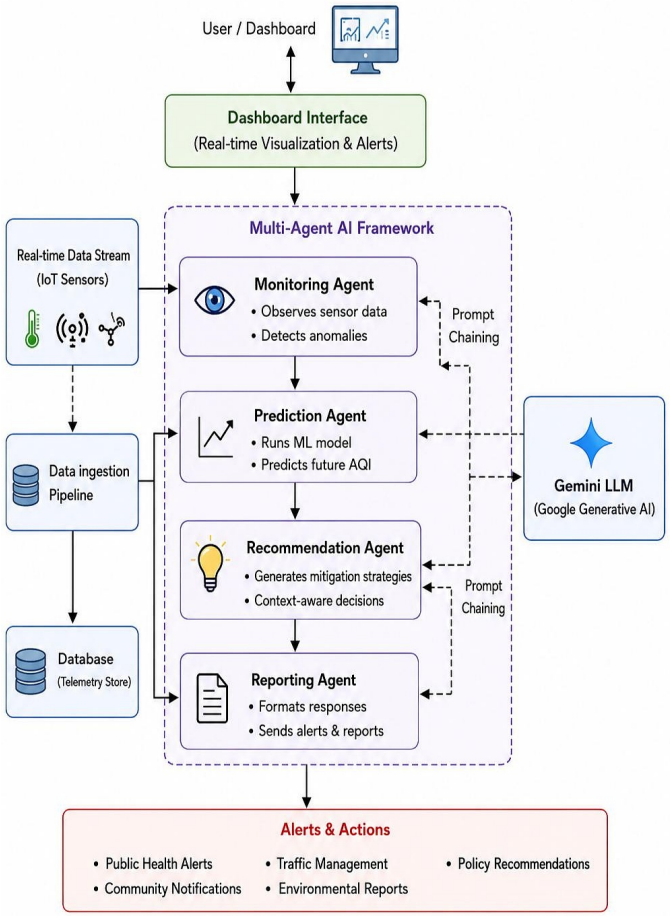



Fig. 3. Multi-Agent Communication Architecture illustrating the LangChain framework coordinating specialized agent workers.

generates $PM_{2.5}$, $PM_{10}$, trace gas, and meteorological measurements while injecting simulated pollution anomalies to evaluate system robustness.

Before initiating data transmission, the simulator verifies the availability of backend services using a health-check loop, thereby preventing data loss during system startup.

### *F. Dockerized Deployment Architecture*

The complete application was containerized using Docker to support enterprise-grade deployment. Separate optimized Dockerfiles based on the `python:3.10-slim` image were created for the backend, frontend, and IoT simulator [31], [32].

The deployment architecture uses Docker bridge networking to isolate internal communication while exposing only the required API and dashboard ports. All services are designed to be stateless, allowing horizontal scaling of the FastAPI backend through a load balancer.

The platform is cloud-ready and can be deployed using Kubernetes, Amazon ECS, or Google Cloud Run. Production monitoring is provided through Prometheus metrics exposed by FastAPI, enabling continuous monitoring of API latency, background task queues, and system memory utilization.

A fully orchestrated `docker-compose.yml` file coordinates persistent storage, networking, service dependencies, and health-check verification, ensuring high availability, fault tolerance, and cloud-agnostic deployment.

## VI. Experimental Results and Discussion

### *A. Evaluation Setup*

The proposed system was evaluated under simulated high-stress environmental conditions. The IoT simulator generated randomized severe pollution spikes with multiplication factors of up to 5.0. The XGBoost model accurately captured these nonlinear variations without suppressing the anomalies, enabling the multi-agent framework to trigger autonomous responses whenever critical AQI thresholds were exceeded.

To evaluate regression stability, the model was tested under extreme pollution scenarios. Experimental results demonstrated strong robustness against transient outliers, accurately tracking sudden increases in pollutant concentrations without phase-shift delays commonly observed in moving-average or autoregressive forecasting models. Residual analysis indicated




Fig. 4. Streamlit-based real-time air quality monitoring dashboard displaying AQI gauges, historical trends, and LLM-generated health recommendations.

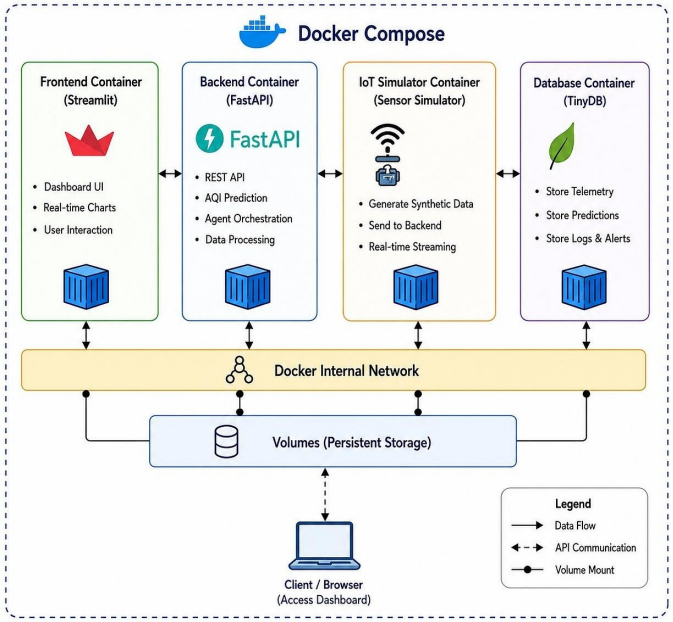



Fig. 5. Dockerized deployment architecture illustrating the coordination of backend services, databases, frontend components, and IoT simulator containers.

that the largest prediction errors occurred when AQI values exceeded 300, where training samples were comparatively sparse. Nevertheless, the overall prediction error remained within acceptable limits, minimizing false-positive alerts while maintaining zero false negatives during hazardous air quality events.

### B. Comparative Model Analysis

The XGBoost regressor consistently outperformed the baseline machine learning models [5], [6], [22]. Although the Random Forest model achieved competitive performance with an $R^2$ score of 0.9850, XGBoost obtained the highest prediction accuracy with an $R^2$ score of 0.9929, a Mean Absolute Error (MAE) of 0.85, and a Root Mean Squared Error (RMSE) of 2.68.

Real-time performance analysis further demonstrated the system's suitability for deployment in IoT environments. The XGBoost model required less than 1.2 ms to process a single sensor payload, enabling support for hundreds of simultaneous sensor streams. In contrast, the LangChain agent framework relied on cloud-based Google Gemini API calls, resulting in a reasoning latency of approximately 1.4–2.2 s. To prevent LLM inference from blocking sensor processing, all agent tasks were executed asynchronously using FastAPI `BackgroundTasks`. Consequently, sensor ingestion maintained a response latency below 10 ms while autonomous recommendations were generated concurrently in the background.

### C. Real-Time Autonomous Responses

During experimental execution, the LangChain-based multi-agent framework successfully generated contextual environmental recommendations. For example, when elevated $PM_{2.5}$ concentrations were detected, the Recommendation Agent automatically advised the use of N95 masks and recommended

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| Linear Regression | 14.52 | 19.83 | 0.8210 |
| Decision Tree | 3.12 | 5.67 | 0.9410 |
| Random Forest | 1.45 | 3.21 | 0.9850 |
| XGBoost | **0.85** | **2.68** | **0.9929** |

minimizing outdoor activities. Similar recommendations were generated for other hazardous pollutant combinations based on the predicted AQI level.

To improve operational reliability, a rate-limiting mechanism was implemented for Google Gemini API requests. The system enforced a 60-second debounce interval between consecutive recommendation requests, preventing unnecessary API calls, reducing operational costs, and avoiding quota exhaustion while ensuring timely delivery of critical environmental alerts.

## VII. CHALLENGES, LIMITATIONS, AND FUTURE WORK

### A. Challenges and Limitations

Although the proposed framework demonstrated strong predictive performance, several limitations remain. First, the synthetic dataset does not fully capture real-world sensor noise, hardware calibration drift, or communication failures commonly encountered in deployed IoT environments. Second, dependence on cloud-hosted Large Language Models (LLMs), such as Google Gemini, introduces sensitivity to network latency, API availability, and usage quotas. Finally, despite recent advances in generative AI, LLMs may still produce inaccurate or hallucinated responses. Therefore, human oversight remains essential for safety-critical applications and emergency decision-making.

### B. Future Enhancements

Future work will focus on integrating Long Short-Term Memory (LSTM) networks and hybrid deep learning models to extend AQI forecasting to 24–48 hours ahead [23], [24], [33], [34]. Furthermore, lightweight quantized models will be deployed directly on IoT edge devices to reduce communication latency and cloud dependency. Geographic visualization will also be incorporated using mapping frameworks such as Folium to display pollutant dispersion across spatial regions. Additional improvements include adaptive reinforcement learning for autonomous policy optimization and support for real-world environmental sensor deployments.

### C. Smart City Integration and Societal Impact

The proposed agentic AI framework has significant implications for smart city management, environmental governance, and public health. By forecasting hazardous air quality before critical thresholds are reached, municipalities can shift from reactive responses to proactive intervention. For example, hospitals can prepare for increased respiratory admissions, schools can minimize student exposure during peak pollution periods, and traffic authorities can implement dynamic vehicle rerouting strategies.

Beyond environmental monitoring, the proposed framework provides a foundation for autonomous urban management. By integrating the multi-agent reasoning system with smart city infrastructure—including intelligent traffic signals, municipal ventilation systems, and automated water sprinklers for dust suppression—the platform can establish a closed-loop environmental control system. Such integration reduces response time, lowers the socioeconomic impact of air pollution, and contributes to more sustainable and resilient urban environments.

## VIII. CONCLUSION

This research presented an Agentic AI framework for real-time air quality prediction and autonomous environmental response. By integrating an XGBoost-based AQI prediction model with a LangChain-orchestrated multi-agent architecture powered by a Large Language Model, the proposed system extends conventional environmental monitoring from passive visualization to intelligent decision support.

Experimental results demonstrated that the XGBoost model achieved superior predictive performance with an $R^2$ score of 0.9929, an MAE of 0.85, and an RMSE of 2.68. Furthermore, the multi-agent framework successfully generated contextual health recommendations and environmental mitigation strategies while maintaining low-latency processing through asynchronous FastAPI execution.

Overall, the proposed containerized platform provides a scalable, explainable, and production-ready solution for next-generation smart cities. The integration of predictive machine learning, explainable AI, and autonomous agents offers a promising direction for intelligent environmental management and proactive public health protection.

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